Chapter 8

Effect of Electrolytes on Chemical Equilibria

***Fundamentals of Analytical Chemistry*, Skoog, et al., 10th Ed., 2022**

Chapter 8 Problems:

8-1, 8-3, 8-9, 8-10

The position of most solution equilibria depends on the electrolyte concentration of the medium, even when the added electrolyte contains no common ion for the reaction.

$$F = \frac{q_1 q_2}{\varepsilon r^2}$$

$$AgCl_{(s)} \leftrightarrow Ag^+_{(aq)} + Cl^-_{(aq)}$$

The Salt Effect (aka electrolyte effect)

Results from the electrostatic attractive and repulsive forces between the ions of an electrolyte and the ions involved in an equilibrium. These forces cause each ion from the dissociated reactant to be surrounded by a sheath of solution that contains a slight excess of electrolyte ions of opposite charge.

The solubility becomes greater as the number of electrolyte ions in the solution becomes larger. In other words, the effective concentration of silver ions and chloride ions becomes less as the ionic strength of the medium becomes greater.

Concentration based equilibrium constants are indicated by adding a prime mark K'_w , K'_{sp}, K'_a

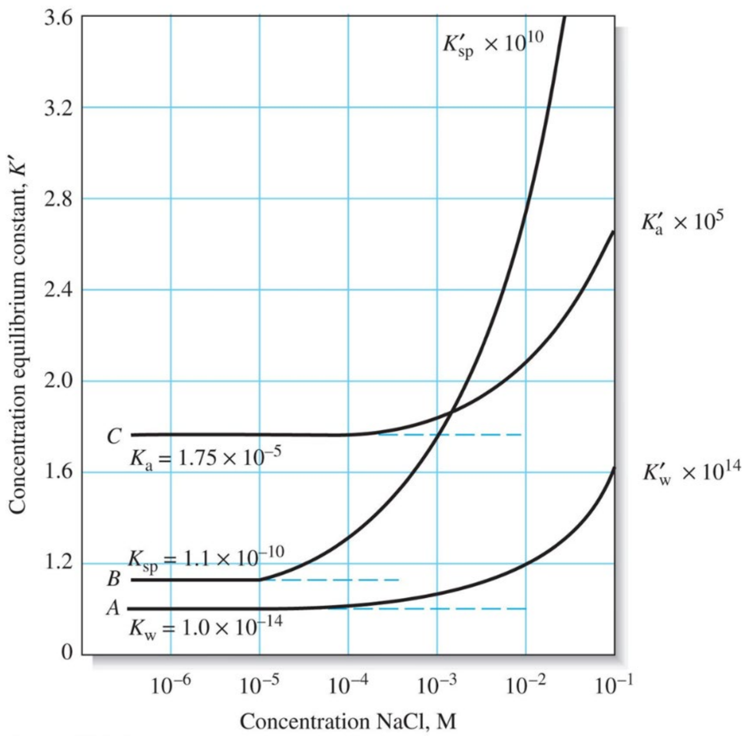
As the electrolyte concentration becomes very small, concentration-based equilibrium constants approach their thermodynamic values: K_w, K_{sp}, K_a

Figure 8-1

Curve A product of molar **hydronium and hydroxide** concentrations

Curve B product of molar **barium and sulfate** concentration in a saturated solution

Curve C concentration-based equilibrium constant for **acetic acid dissociation**



The electrolyte effect shown in Figure 8-1 is not unique to sodium chloride, we see nearly identical curves if potassium nitrate or sodium perchlorate are used instead of sodium chloride.

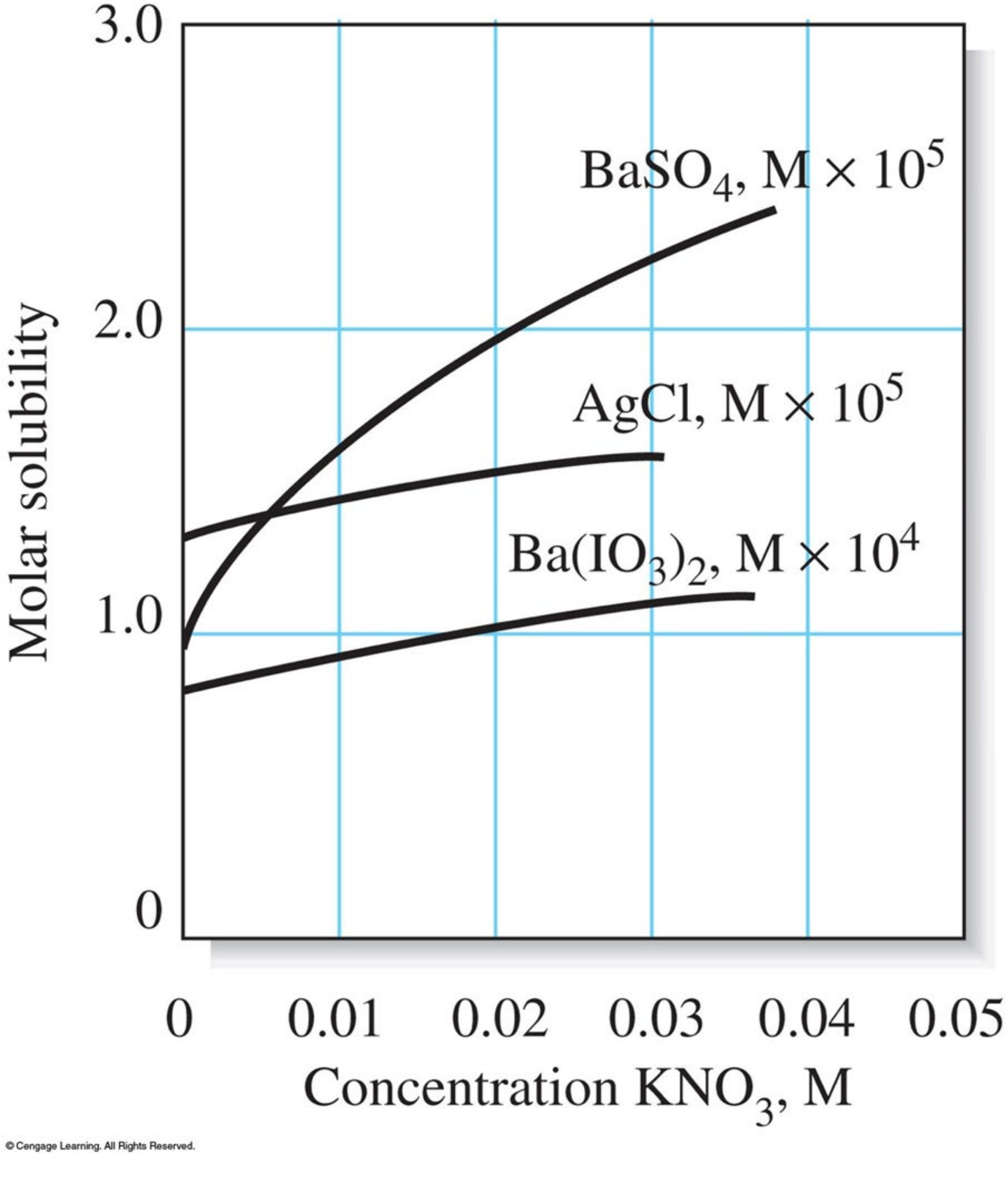
The origin of the effect is the electrostatic attraction between the ions of the electrolyte and the ions of the reacting species of opposite charge. Since the electrostatic forces associated with all singly charged ions are approximately the same, the three salts exhibit essentially the same effect.

Ions of the same charge have essentially the same effect on the equilibria.

The Effect of Ionic Charges on Equilibria

The magnitude of the electrolyte effect is highly dependent of the charges of the participants in an equilibrium.

Figure 8-2



The Effect of Ionic Strength

The effect of added electrolyte on equilibria is independent of the chemical nature of the electrolyte but depends on a property of the solution called the ionic strength.

[X] represents the molar concentration

Z_x represents the charge of the species

$$\text{Ionic Strength} = \mu = \tfrac{1}{2}\left([A]Z_A^2 + [B]Z_B^2 + [C]Z_C^2 + \ldots\right)$$

Example 8-1

1. **Calculate the ionic strength of (a) 0.1 M solution of KNO_3 (b) 0.1 M solution of Na_2SO_4**

Example 8-2

2. **What is the ionic strength of solution that is 0.05 M KNO_3 and 0.1 M Na_2SO_4**

For solutions with ionic strengths of 0.1 M or less, the electrolyte effect is independent of the kind of ions and dependent only on the ionic strength.

Activity Coefficients

The activity of a species is a measure of its effective concentration. It depends on the ionic strength of the medium.

a_x = activity or effective concentration

γ_x = activity coefficient

$$a_x = [X]\,\gamma_x$$

If we substitute a_x for [X] in any equilibrium constant expression we find the equilibrium constant is then independent of ionic strength.

Thermodynamic

$$k_{sp} = a_x^m a_y^n$$

$$k_{sp} = [X]^m \gamma^m [Y]^n \gamma^n$$

Concentration

$$k'_{sp} = [X]^m [Y]^n$$

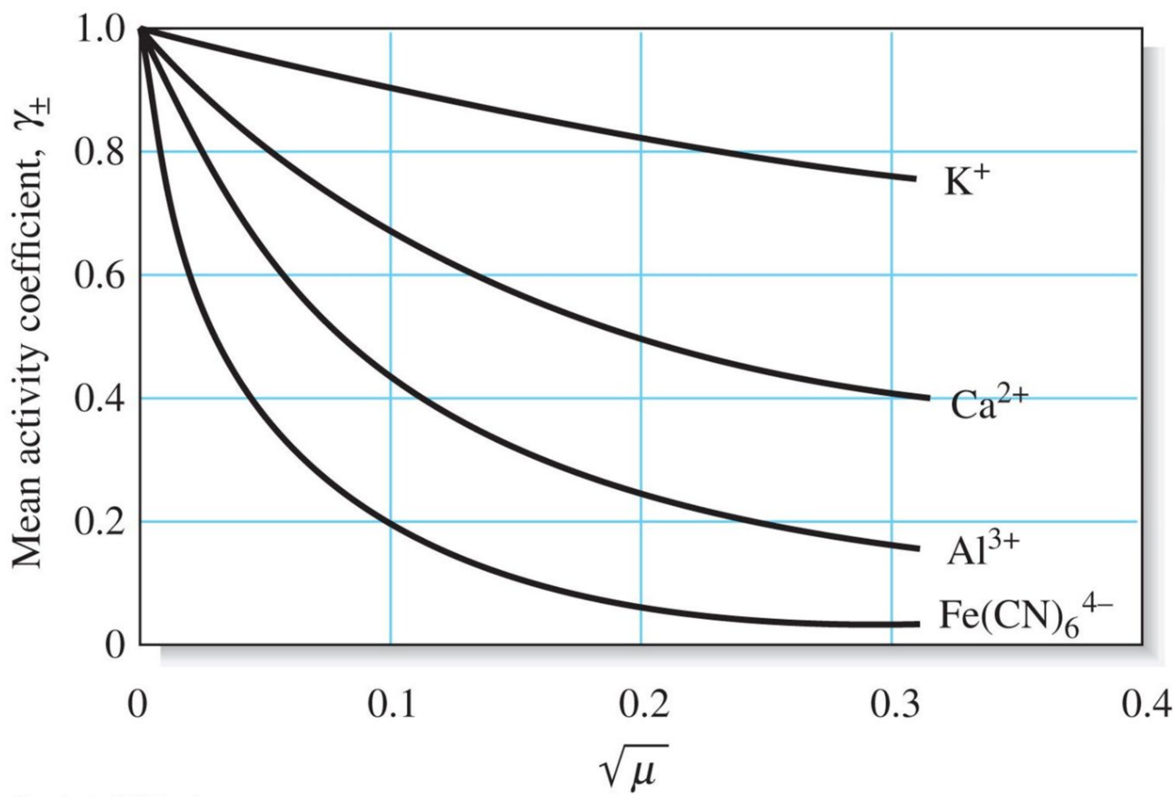
Therefore

$$k_{sp} = k'_{sp} \gamma^m \gamma^n$$

Properties of Activity Coefficients

1. **The activity coefficient depends on the solution ionic strength**
2. **In very dilute solutions, the activity coefficient approaches unity**
3. **For a given ionic strength, the activity coefficient gets smaller as the charge of the chemical species increases.** ***(Figure 10-3)***
4. **The activity coefficient of an uncharged molecule is approximately unity, no matter what the level of ionic strength.**
5. **At any ionic strength the activity coefficients are approximately equal for species having the same charge.**

Fig. 10-3



The Debye-Huckel Equation

$$-\log \gamma_x = \frac{0.51 * {Z_x}^2 * \sqrt{\mu}}{1 + (3.3 * \alpha_x * \sqrt{\mu}\,)}$$

The constants 0.51 and 3.3 are applicable to aqueous solutions at 25°C.

Example 8-3

3. **Use the Debye-Huckel Equation to calculate the activity coefficient for H_g^{2+} in a solution that has an ionic strength of 0.085 M. Lookup the effective diameter α_x in table 10-2**

TABLE 10-2

Activity Coefficients for Ions at 25°C

		Activity Coefficient at Indicated Ionic Strength				
Ion	**α_X, nm**	**0.001**	**0.005**	**0.01**	**0.05**	**0.1**
H_3O^+	0.9	0.967	0.934	0.913	0.85	0.83
Li^+, $C_6H_5COO^-$	0.6	0.966	0.930	0.907	0.83	0.80
Na^+, IO_3^-, HSO_3^-, HCO_3^-, $H_2PO_4^-$, $H_2AsO_4^-$, OAc^-	0.4–0.45	0.965	0.927	0.902	0.82	0.77
OH^-, F^-, SCN^-, HS^-, ClO_3^-, ClO_4^-, BrO_3^-, IO_3^-, MnO_4^-	0.35	0.965	0.926	0.900	0.81	0.76
K^+, Cl^-, Br^-, I^-, CN^-, NO_2^-, NO_3^-, $HCOO^-$	0.3	0.965	0.925	0.899	0.81	0.75
Rb^+, Cs^+, Tl^+, Ag^+, NH_4^+	0.25	0.965	0.925	0.897	0.80	0.75
Mg^{2+}, Be^{2+}	0.8	0.872	0.756	0.690	0.52	0.44
Ca^{2+}, Cu^{2+}, Zn^{2+}, Sn^{2+}, Mn^{2+}, Fe^{2+}, Ni^{2+}, Co^{2+}, Phthalate^{2-}	0.6	0.870	0.748	0.676	0.48	0.40
Sr^{2+}, Ba^{2+}, Cd^{2+}, Hg^{2+}, S^{2-}	0.5	0.869	0.743	0.668	0.46	0.38
Pb^{2+}, CO_3^{2-}, SO_3^{2-}, $C_2O_4^{2-}$	0.45	0.868	0.741	0.665	0.45	0.36
Hg_2^{2+}, SO_4^{2-}, $S_2O_3^{2-}$, Cr_4^{2-}, HPO_4^{2-}	0.40	0.867	0.738	0.661	0.44	0.35
Al^{3+}, Fe^{3+}, Cr^{3+}, La^{3+}, Ce^{3+}	0.9	0.737	0.540	0.443	0.24	0.18
PO_4^{3-}, $Fe(CN)_6^{3-}$	0.4	0.726	0.505	0.394	0.16	0.095
Th^{4+}, Zr^{4+}, Ce^{4+}, Sn^{4+}	1.1	0.587	0.348	0.252	0.10	0.063
$Fe(CN)_6^{4-}$	0.5	0.569	0.305	0.200	0.047	0.020

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